PRINCIPLES OF DEEP NETURAL COMPUTATION

Demand Forecasting for Retail Stores

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PROBLEM STATEMENT

Retail stores often face difficulty in predicting how much of each product customers will buy.

Because of this:

Too much stock (Overstocking): Leads to wastage, extra storage cost, and unsold products

Too little stock (Stock shortage): Leads to lost sales and unhappy customers

Demand for products keeps changing due to:

- Seasons (summer, winter)
- Festivals and holidays
- Discounts and offers

Without proper prediction, retailers make decisions based on guesswork, which is not reliable.

ABSTRACT

Demand forecasting plays an important role in retail by helping predict future product demand and manage inventory effectively. Incorrect predictions can lead to overstocking and stock shortages, affecting business performance.

This project uses historical sales data, including product details, sales quantity, dates, promotions, and holidays. Data preprocessing and feature engineering are applied to improve data quality and capture important patterns like seasonality.

A hybrid model combining time series analysis and LSTM deep learning is used for accurate prediction. The model is evaluated using MAE and RMSE metrics.

The system helps improve inventory management, reduce costs, and increase customer satisfaction.

Objectives:

- Predict future product demand using historical sales data
- Improve inventory management by maintaining optimal stock levels
- Reduce overstocking to minimize storage cost and wastage
- Avoid stock shortages to prevent loss of sales
- Analyze demand patterns such as seasonal trends and promotions
- Apply feature engineering to extract meaningful insights from data
- Develop a hybrid forecasting model using time series analysis and LSTM
- Train the model to learn past sales behavior accurately
- Evaluate model performance using MAE and RMSE metrics
- Provide accurate and reliable demand predictions
- Support better decision-making for retailers
- Improve overall business efficiency and customer satisfaction

Literature Survey

S.No	Author & Year	Method Used	Key Idea / Contribution	Limitations
1	Smith et al. (2018)	Time Series (ARIMA)	Used statistical methods to forecast retail demand based on past sales trends	Cannot handle complex and non-linear patterns
2	Lee & Chen (2019)	Machine Learning (Regression)	Applied regression models for demand prediction using multiple features	Limited accuracy for large and complex datasets
3	Kumar et al. (2020)	Random Forest	Used ensemble learning to improve prediction accuracy	Requires high computational resources
4	Zhang et al. (2021)	Deep Learning (LSTM)	Captured long-term dependencies in time-series data for better forecasting	Needs large dataset and training time
5	Patel & Shah (2022)	Hybrid Model (ARIMA + LSTM)	Combined statistical and deep learning models for improved accuracy	Increased model complexity

SYSTEM WORKFLOW

Step	Description
1. Data Collection	Gather historical retail sales data including Product ID, sales quantity, date of sale, promotions, and holidays.
2. Data Preprocessing	Remove duplicates, handle missing values, convert dates into time-series format, and normalize numerical values.
3. Feature Engineering	Extract important features such as historical sales patterns, seasonal trends, promotions, holiday effects, and product categories.
4. Hybrid Model Development	Combine Time Series Analysis and LSTM Deep Learning to capture trends, seasonality, and complex demand patterns.
5. Model Training	Split data into training and testing sets, train the model on historical patterns, and validate performance using testing data.
6. Demand Prediction	Predict daily, weekly, and monthly demand to help maintain optimal inventory levels.
7. Performance Evaluation	Evaluate model accuracy using MAE and RMSE metrics and identify areas for improvement.
8. Deployment / Accessibility	Implement using Python, TensorFlow/Keras, Pandas; share model via GitHub; applicable to various retail environments.

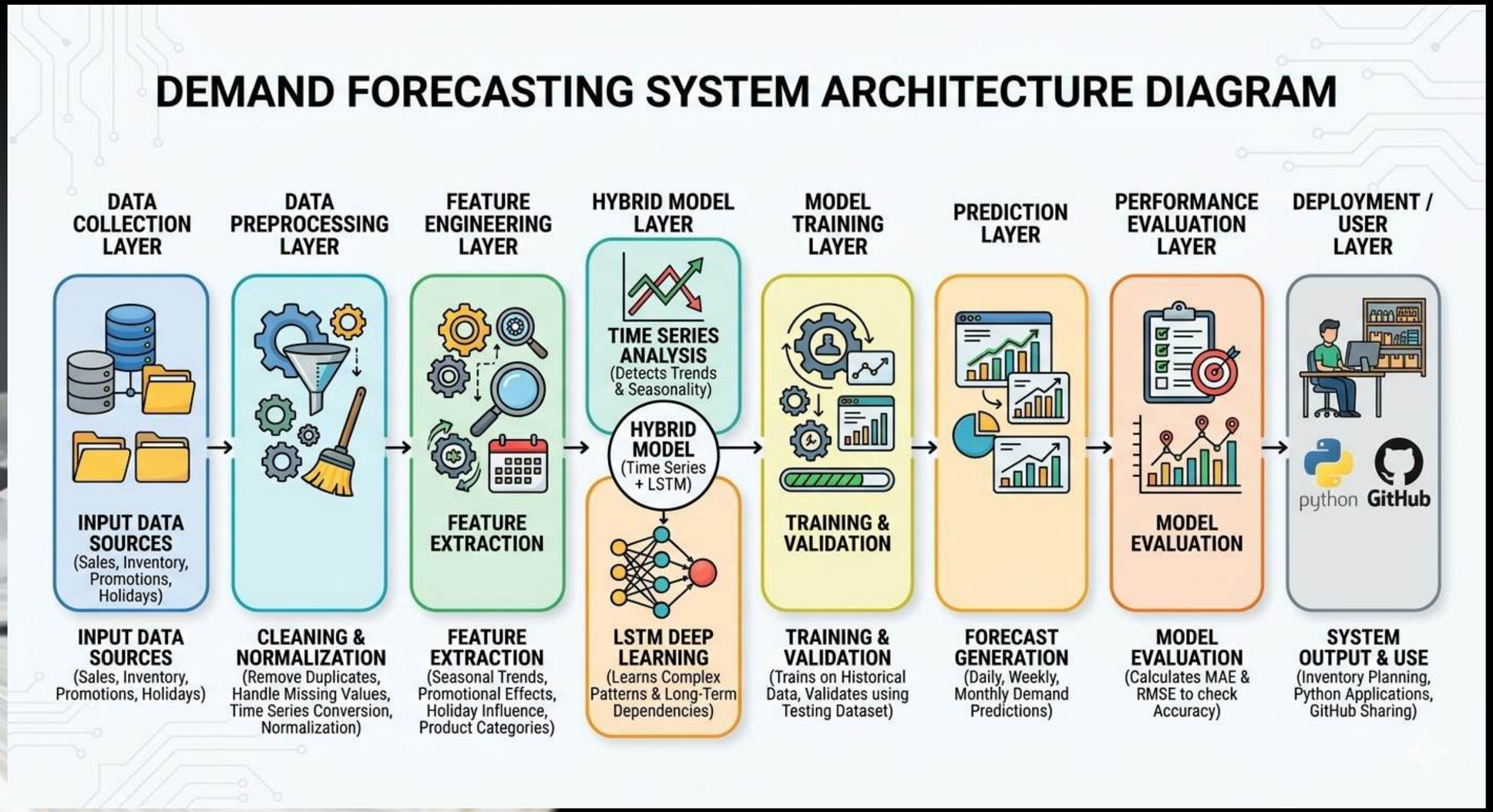
Existing Systems

- Mostly use traditional statistical models like ARIMA, Moving Average, or Regression.
- Can predict simple demand patterns based on historical data.
- Limited in handling complex and non-linear demand patterns.
- Struggle with seasonal trends, promotions, and holiday effects.
- Accuracy is lower for large datasets or long-term forecasting.
- Difficult to adapt to dynamic retail environments.

PROPOSED SYSTEM

- Uses a hybrid model combining Time Series Analysis and LSTM deep learning.
- Can capture trends, seasonality, and complex demand relationships.
- Incorporates feature engineering: promotions, holidays, product categories.
- Provides accurate daily, weekly, and monthly demand predictions.
- Improves inventory management, reduces wastage, and prevents stock shortages.
- Adaptable to different retail environments, including supermarkets and online stores.

ARCHITECTURE DIAGRAM



SAMPLE OUTPUTS

Input Data

Product ID	Date	Sales Quantity	Promotion	Holiday
P001	1/3/2026	50	Yes	No
P002	1/3/2026	30	No	No
P003	1/3/2026	70	Yes	Yes

Predicted Demand

Product ID	Predicted Daily Demand	Predicted Weekly Demand	Predicted Monthly Demand
P001	55	380	1650
P002	32	224	960
P003	75	525	2250

Performance Metrics

Metric	Value
MAE	3.5
RMSE	5.2

SAMPLE CODE

```
# Sample Demand Forecasting using LSTM

# Import libraries
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

# Step 1: Load sample data
data = {
    'Date': pd.date_range(start='2026-03-01', periods=10, freq='D'),
    'Sales': [50, 55, 53, 60, 58, 65, 63, 70, 68, 75]
}
df = pd.DataFrame(data)
df.set_index('Date', inplace=True)

# Step 2: Normalize sales data
scaler = MinMaxScaler(feature_range=(0,1))
scaled_data = scaler.fit_transform(df)

# Step 3: Prepare data for LSTM
X = []
y = []
for i in range(3, len(scaled_data)):
    X.append(scaled_data[i-3:i, 0])  # past 3 days
    y.append(scaled_data[i, 0])
X, y = np.array(X), np.array(y)
X = X.reshape((X.shape[0], X.shape[1], 1))  # LSTM input shape
```

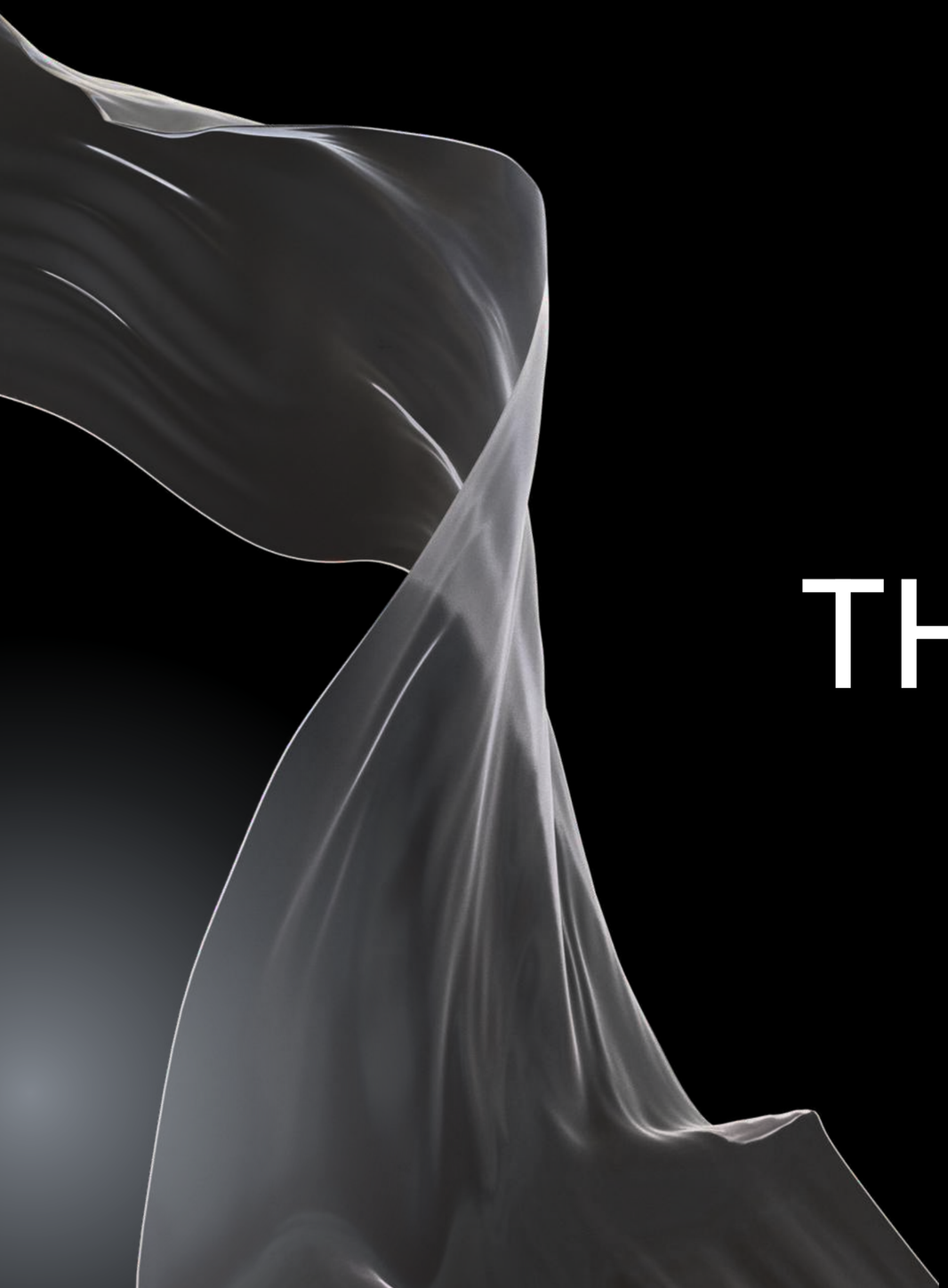
```
# Step 4: Build LSTM model
model = Sequential()
model.add(LSTM(50, activation='relu', input_shape=(3,1)))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mse')

# Step 5: Train model
model.fit(X, y, epochs=100, verbose=0)

# Step 6: Predict next day's demand
last_3_days = scaled_data[-3:].reshape((1,3,1))
predicted_scaled = model.predict(last_3_days)
predicted_demand = scaler.inverse_transform(predicted_scaled)
print("Predicted next day's sales:", predicted_demand[0][0])
```

OUTPUT

Predicted next day's sales: 77.3

CONCLUSION

- The proposed demand forecasting system helps retailers predict future product demand accurately.
- Combining time series analysis with LSTM deep learning improves prediction accuracy over traditional methods.
- The system reduces overstocking and stock shortages, leading to better inventory management.
- Retailers can plan inventory efficiently, reduce costs, and improve customer satisfaction.
- The methodology is flexible and can be applied to different retail environments, including supermarkets and online stores.
- Overall, the system supports data-driven decision making and enhances business performance.



THANK YOU